PROJECT 3: COMPARISON OF DEEP LEARNING ARCHITECTURES

Submitted By

Name: Aarush C S

Reg.No: 2411021240028

1. Introduction

Deep Learning is a branch of Machine Learning that uses artificial neural networks with multiple layers to learn patterns and features from data. It has become an important technology in areas such as computer vision, natural language processing, speech recognition, recommendation systems, and time-series prediction.

Different types of data require different neural network architectures. A model that works well with tabular data may not be the best choice for images or sequential information. Three fundamental architectures used in deep learning are **Feedforward Neural Networks (FNNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs)**.

Feedforward Neural Networks process information in a single forward direction and are commonly used for structured data. CNNs are designed to identify spatial patterns and are widely used for image-related tasks. RNNs are designed to work with sequential data where the order of information is important.

This project studies the basic architecture, working principles, suitable applications, and differences between these three architectures.

2. Objectives

The main objectives of this project are:

- To study the basic architecture of Feedforward Neural Networks, CNNs, and RNNs.
- To understand how each architecture processes information.
- To identify suitable applications for each architecture.
- To compare the three architectures based on their structure and input data.
- To understand how to select a suitable architecture for a particular deep learning problem.

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3. Feedforward Neural Networks (FNN)

3.1 Basic Architecture

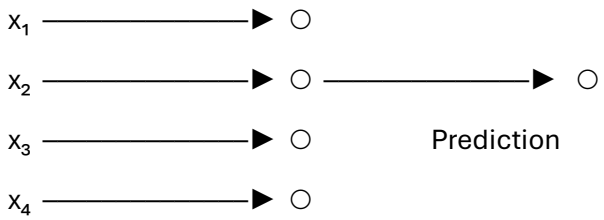
A **Feedforward Neural Network (FNN)** is one of the simplest types of artificial neural networks. In a feedforward network, information flows in only one direction, from the input layer to the hidden layers and finally to the output layer.

A basic FNN consists of:

Input Layer → Hidden Layer(s) → Output Layer

Architecture

Input Layer Hidden Layers Output Layer



The **input layer** receives the features of the dataset. The **hidden layers** perform calculations using weights, biases, and activation functions. The **output layer** produces the final prediction.

3.2 How FNN Works

Each neuron receives input values and calculates a weighted sum. An activation function is then applied to introduce non-linearity.

During training, the network compares its prediction with the actual result. The error is propagated backward through the network using **backpropagation**, and the weights are adjusted to improve future predictions.

FNNs do not have a memory mechanism. Therefore, they treat each input independently and do not naturally understand spatial or sequential relationships.

3.3 Applications of FNN

Feedforward networks are suitable for problems where the input consists mainly of structured or numerical features.

Common applications include:

- **Classification:** Predicting categories such as customer types.
- **Regression:** Predicting numerical values such as prices.
- **Customer Analysis:** Classifying customers based on their characteristics.
- **Financial Prediction:** Analyzing structured financial information.
- **Tabular Data:** Working with datasets containing rows and features.

Example

Suppose a model needs to predict whether a student will pass an examination based on:

- Attendance
- Assignment marks
- Internal marks
- Study hours

An FNN can use these values as input and predict **Pass** or **Fail**.

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4. Convolutional Neural Networks (CNN)

4.1 Basic Architecture

A **Convolutional Neural Network (CNN)** is a deep learning architecture mainly designed for processing images and other data with spatial relationships.

A typical CNN consists of:

Input → Convolution → Activation → Pooling → Fully Connected Layer → Output

Architecture

Input Image

Convolution Layer

Feature Maps

Pooling Layer

Fully Connected Layer



Output / Prediction

The main feature of CNNs is the **convolution layer**, which uses filters to detect important patterns in an image.

4.2 How CNN Works

An image contains pixels arranged in a spatial structure. Instead of treating every pixel independently, CNNs examine small regions of the image.

The convolution filters can learn to detect simple features such as:

- Edges

- Lines
- Corners
- Textures

As the data moves through deeper layers, the network can combine these simple features to recognize more complex patterns such as shapes, objects, and faces.

The **pooling layer** reduces the size of the feature maps while retaining important information. This helps reduce computational requirements.

Finally, the extracted features are passed to fully connected layers, which produce the final prediction.

4.3 Applications of CNN

CNNs are widely used in **Computer Vision**.

Common applications include:

- **Image Classification:** Identifying what an image contains.
- **Object Detection:** Detecting and locating objects in images.
- **Face Recognition:** Identifying or verifying faces.
- **Medical Image Analysis:** Analyzing medical scans and images.
- **Autonomous Vehicles:** Detecting objects such as vehicles, pedestrians, and road signs.
- **Quality Inspection:** Detecting defects in manufactured products.

Example

Consider a system that identifies whether an image contains a **cat or a dog**.

The CNN receives the image as input and learns visual features such as edges, shapes, textures, and body structures. After processing these features, the output layer predicts whether the image represents a cat or a dog.

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5. Recurrent Neural Networks (RNN)

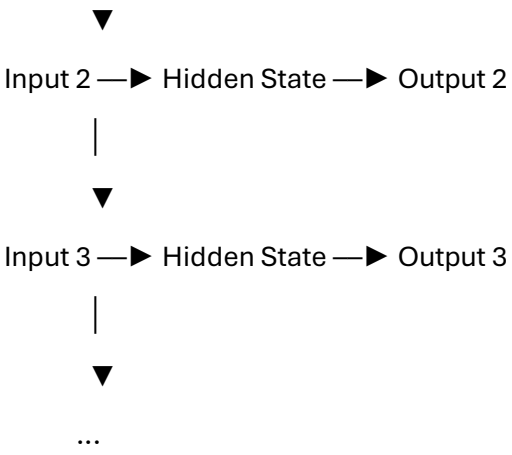
5.1 Basic Architecture

A **Recurrent Neural Network (RNN)** is designed for processing **sequential data**. In sequential data, the order of the information is important.

Unlike a basic feedforward network, an RNN maintains information from previous inputs through a hidden state.

Architecture

Input 1 —► Hidden State —► Output 1



The hidden state acts as a form of memory, allowing information from previous time steps to influence the processing of the current input.

5.2 How RNN Works

An RNN processes data one element at a time. At each step, it receives:

1. The current input.
2. The hidden state from the previous step.

These are used to calculate a new hidden state.

For example, when processing a sentence, the meaning of the current word can depend on words that appeared earlier in the sentence. An RNN attempts to maintain relevant information from previous words while processing the next word.

RNNs are therefore useful for data where **sequence and time order matter**.

Traditional RNNs can have difficulty remembering information over very long sequences. This led to improved architectures such as **Long Short-Term Memory (LSTM)** and **Gated Recurrent Unit (GRU)** networks.

5.3 Applications of RNN

RNNs are commonly used for sequential and time-dependent problems.

Applications include:

- **Text Processing:** Processing sequences of words.
- **Speech Recognition:** Processing audio signals over time.
- **Language Modeling:** Predicting the next word in a sequence.
- **Time-Series Forecasting:** Predicting future values from historical data.
- **Sentiment Analysis:** Analyzing the sequence of words in a sentence.
- **Sequence Classification:** Classifying sequential data based on its patterns.

Example

Consider predicting the next word in:

"I am going to the ___."

The model processes the words in sequence. Information from the previous words helps it predict a suitable next word such as **"college"** or **"market"**.

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6. Comparison of FNN, CNN and RNN

Feature	FNN	CNN	RNN
Full Name	Feedforward Neural Network	Convolutional Neural Network	Recurrent Neural Network
Basic Structure	Input → Hidden → Output	Convolution → Pooling → Fully Connected	Sequential inputs with hidden state
Main Input	Structured / tabular data	Images / spatial data	Sequential / time-series data
Information Flow	Forward direction	Layer-by-layer feature extraction	Sequential with recurrent connections
Memory	No memory	No traditional recurrent memory	Maintains hidden state
Main Strength	Feature-based learning	Spatial feature extraction	Sequence processing
Spatial Understanding	Low	High	Low
Sequence Understanding	Low	Low	High
Common Applications	Classification, regression	Image classification, object detection	Text, speech, time-series
Example	Student performance prediction	Cat/dog image classification	Next-word prediction

6.1 Key Differences

The most important difference between the three architectures is the type of information they are designed to process.

FNN:
FNNs are suitable when the input is a set of independent features and there is no important spatial or sequential relationship between them.

CNN:
CNNs are suitable when the relationship between nearby data points is important. They are

particularly effective for images because pixels close to one another contain useful spatial information.

RNN:
RNNs are suitable when the order of the input matters. They can use information from previous steps when processing the current step.

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7. Choosing the Right Architecture

The selection of a deep learning architecture depends mainly on the type of input data and the problem being solved.

For structured data → FNN

If the dataset consists mainly of numerical or structured features, an FNN can be a suitable choice.

Example:
Predicting house prices using area, number of rooms, location and age.

For images → CNN

If the problem involves images or spatial patterns, CNNs are generally more suitable.

Example:
Detecting whether an X-ray image shows signs of a particular condition.

For sequential data → RNN

If the order of data is important, an RNN can be suitable.

Example:
Predicting future sales using sales data from previous months.

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8. Conclusion

Feedforward Neural Networks, Convolutional Neural Networks, and Recurrent Neural Networks are important deep learning architectures, but each one is designed for different types of problems.

FNNs provide a simple approach for structured data and are commonly used for classification and regression. **CNNs** are specialized for spatial data and have become highly important in computer vision applications such as image classification and object detection. **RNNs** are designed for sequential data and can use information from previous time steps, making them useful for text, speech, and time-series applications.

Therefore, understanding the differences between these architectures is important when designing a deep learning solution. The appropriate architecture should be selected based on the **structure of the data, the relationships within the data, and the objective of the application**.

Expected Learning Outcome

After completing this project, students will be able to:

- Explain the basic architecture of FNNs, CNNs, and RNNs.
- Differentiate between the three architectures.
- Identify the type of input data suitable for each architecture.
- Explain common applications of each architecture.
- Select an appropriate deep learning architecture for a basic problem.